Module 17

PROCESS DOCUMENTATION

1. Information	
Process Name:	Supply Inventory Tracking Process
Process Owner:	Clinic Doctor / Clinic Owner
Department/Unit:	Inventory, Supplies & Operations
Document Prepared By:	Tristan Paolo C. De Castro
Date Prepared:	September 12, 2026
Version Number:	1.0
Last Updated:	September 12, 2026

2. Overview

Purpose: To replace the current reactive method — noticing supplies are low only while using them — with zero manual counting required.

Process Description: The system maintains a master list of supplies with quantity on hand. When the doctor charts a procedure and taps supplies used, the system auto-deducts them. Real-time quantity is visible on the dashboard, and full deduction history is retained.

Actors Involved:
- Doctor — taps supplies used during charting
- System — deducts, tracks, displays

3. Inputs

- Supplies Used - Module 9 (Dental Digital Charting)
- Master Supply List - Module 17 config

4. Activities

1.System maintains master supply list - System
2. Doctor taps supplies used during charting - Doctor
3. System auto-deducts supplies - System
4. System updates real-time quantity - System
5. System keeps deduction history - System
6. Doctor views live count on dashboard - Doctor

5. Outputs

- Live Stock Count -> Doctor (Module 21: Admin & Operations Dashboard)
- Deduction History -> Module 17
- Low-Stock Trigger -> Module 18 (Low-Stock Alert)

6. Stakeholders Analysis

Internal:
- Doctor / Clinic Owner — views stock, uses supplies
- IT Department — maintains inventory system

External:
- Suppliers — receive orders (via Module 20)

Note: *No receptionist is involved. Stock deducts automatically during charting.*

7. Flow Summary

Start → Doctor Charts Procedure → Tap Supplies Used → System Auto-Deducts → Update Live Count → End

8. Performance Metrics	
Stock counting	Manual, periodic (Target: Real-time)
Deduction logging	Inconsistent (Target: Automatic)
Stockout incidents	Common (Target: Rare)
Audit time	Hours (Target: Zero)

9. Issues and Challenges

- Supplies noticed as low only during use
- No real-time count
- Manual counting is periodic and error-prone
- No deduction history

10. Improvement Opportunities

- Real-time count
- Auto-deduction via charting - with Module 9 (Dental Digital Charting)
- Deduction history
- Dashboard view - with Module 21 (Admin & Operations Dashboard)

11. Automation Potential

- Deduction - Automatic
- Count update - Automatic
- History logging - Automatic

Only remaining human touchpoint: Doctor taps supplies used (one action).

12. Business Process Lifecycle Analysis	
Design	Create a master supply list with real-time deduction.
Execution	Doctor taps supplies; the system deducts automatically.
Monitoring	Track usage patterns and stockout frequency.
Analysis	Identify high-use supplies and waste.
Optimization	Adjust thresholds and reorder points.

13. Evaluation and Recommendation

Manual supply tracking is reactive and error-prone. Module 17 makes it automatic — one tap during charting keeps everything accurate.

Recommendation:

- Implement Module 17 alongside Module 9 (Dental Digital Charting**)**— they work together.
- Pre-load common supplies per procedure — reduces taps.
- Make the count visible on the dashboard — always know what's low.

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Module 18

Module 18: Low-Stock, Critical Low-Stock, and Overstock Alert Process

1. Information

Field	Details
Process Name	Low-Stock, Critical Low-Stock, and Overstock Alert Process
Process Owner	Clinic Doctor / Clinic Owner
Department/Unit	Inventory, Supplies & Operations
Document Prepared By	Tristan Paolo C. De Castro
Date Prepared	September 12, 2026
Version Number	2.1
Last Updated	September 25, 2026

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2. Overview

Purpose: To give advance, severity-based warning for **critical low stock**, **low stock**, and **overstock**, and to automatically trigger ordering for Critical Low and Low conditions — eliminating manual approval and mid-use discovery of shortages.

Process Description: The system monitors stock levels from Module 17 (Supply Inventory Tracking). It compares the live quantity against three configured thresholds per item: **Critical Low**, **Low/Reorder**, and **Overstock/Max**. When a threshold is breached, the system flags the item, notifies the doctor via the dashboard, and:

- **Critical Low** and **Low** → automatically trigger Module 20 to draft and dispatch an order (end-of-day batch, or immediate for Critical if configured).
- **Overstock** → suppress ordering and create a review note for the doctor.

Actors Involved:

- **System** – monitors, compares, classifies severity, flags, notifies, triggers Module 20
- **Doctor** – views alerts, reviews Overstock notes, handles exceptions

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Threshold and Alert Definitions

Threshold	Condition	Alert Type	Severity	Automatic Action

Critical Low	Qty ≤ Critical Low threshold	Critical Low-Stock Alert	Urgent / Red	Auto-trigger Module 20 (immediate or end-of-day batch)
Low / Reorder	Qty ≤ Reorder threshold but > Critical Low	Low-Stock Alert	Warning / Yellow	Auto-trigger Module 20 (end-of-day batch)
Overstock	Qty ≥ Overstock threshold	Overstock Alert	Informational / Blue	Suppress ordering; log review note for doctor

If multiple thresholds are breached, the highest severity wins: **Critical Low > Low > Overstock**.

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3. Inputs

- **Live Stock Quantity** – Module 17 (Supply Inventory Tracking)
- **Threshold per Item** – Module 18 config (Critical Low, Low/Reorder, Overstock/Max)
- **Pending / Queued Orders** – Module 20 (to avoid duplicate orders)
- **Expiration Dates** – Module 19 (for overstock expiry risk)
- **Usage Trends** – Module 9 (Dental Digital Charting) and Module 17

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4. Activities

1. System maintains threshold configuration per item — Critical Low, Low/Reorder, Overstock/Max — **System / Doctor**
2. System monitors live stock levels — **System**
3. System compares quantity against all configured thresholds — **System**
4. If quantity ≤ Critical Low threshold, system flags item as **Critical**, sends **Critical Low-Stock Alert**, and **auto-triggers Module 20** — **System**
5. Else if quantity ≤ Reorder threshold, system flags item as **Low**, sends **Low-Stock Alert**, and **auto-triggers Module 20** — **System**
6. Else if quantity ≥ Overstock threshold, system flags item as **Overstock**, sends **Overstock Alert**, and **suppresses ordering** — **System**
7. System displays alerts on dashboard by severity — **System**
8. Doctor views alert; no approval needed for ordering — **Doctor**
9. Doctor reviews Overstock note and handles exceptions — **Doctor**
10. System logs alert and action in audit trail — **System**

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5. Outputs

- **Critical Low-Stock Alert** → Doctor (Module 21) + **Auto-Trigger** → **Module 20**

- **Low-Stock Alert** → Doctor (Module 21) + **Auto-Trigger** → **Module 20**
- **Overstock Alert** → Doctor (Module 21) for review; **ordering suppressed**
- **Flagged Item** → Module 20 (Critical/Low only)
- **Overstock Review Note** → Doctor (Module 21)
- **Alert/Action Audit Log** → System audit trail

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6. Stakeholders Analysis

Internal:

- **Doctor / Clinic Owner –** views alerts, reviews Overstock notes, handles exceptions
- **IT Department –** maintains alert logic and threshold configuration

External:

- **Suppliers** – receive orders automatically after Module 20 dispatch

Note: No receptionist is involved. Alerts are automatic, severity-based, and directly trigger ordering for Critical/Low.

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7. Flow Summary

Start → Monitor Stock → Compare to Critical Low, Low/Reorder, and Overstock Thresholds →

- Below Critical? → Yes → Critical Low-Stock Alert + **Auto-Trigger Module 20** → End
- No → Below Reorder? → Yes → Low-Stock Alert + **Auto-Trigger Module 20** → End
- No → Above Overstock? → Yes → Overstock Alert + **Suppress Ordering + Review Note** → End
- No → No Alert → End

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8. Performance Metrics

Metric	Current	Target
Stockout incidents	Common	Rare / 0
Critical stockout response	Not tracked	Immediate / same-day
Alert timing	During use – too late	In advance
Notification method	Manual	Automatic, severity-based
Ordering trigger	Manual	Automatic

Overstock incidents	Not tracked	Rare / 0
Overstock visibility	None	Real-time alert
Threshold accuracy	Manual / ad hoc	Reviewed periodically

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9. Issues and Challenges

- Low stock noticed only during use
- No advance warning
- No threshold monitoring
- No distinction between low and critical stock
- No overstock warning (excess inventory, tied-up capital, expiry risk)
- No link between pending orders and overstock
- Single-threshold logic may over-order or under-order
- Manual ordering creates delays

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10. Improvement Opportunities

- Multi-threshold monitoring: Critical Low, Low/Reorder, Overstock
- Severity-based auto-notification
- Automatic ordering trigger for Critical/Low
- Overstock suppression and review
- Dashboard visibility with color-coded alerts
- Integration with Module 19 (Expiration) and Module 20 (Ordering)

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11. Automation Potential

- Monitoring – **Automatic**
- Threshold comparison – **Automatic**
- Severity classification – **Automatic**
- Flagging – **Automatic**
- Notification – **Automatic**
- Ordering trigger – **Automatic**

Only remaining human touchpoint: Doctor views alerts, reviews Overstock notes, handles exceptions.

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12. Business Process Lifecycle Analysis

Phase	Description

Design	Set Critical Low, Reorder, and Overstock thresholds per item
Execution	System flags, notifies, and triggers Module 20 automatically
Monitoring	Track alert frequency, accuracy, stockout/overstock incidents
Analysis	Identify frequently flagged items and causes
Optimization	Adjust thresholds based on usage, lead time, seasonality, expiry

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13. Evaluation and Recommendation

Module 18 now gives advance warning for both shortages and excess, and directly triggers automatic ordering for Critical/Low conditions. **Critical alerts** trigger urgent replenishment, **low-stock alerts** trigger standard replenishment, and **overstock alerts** suppress ordering and prompt review.

Recommendations:

- Implement Module 18 after Module 17 – needs stock data.
- Set thresholds carefully: **Critical Low < Reorder < Overstock**.
- Use color-coded dashboard alerts for immediate visibility.
- Auto-trigger Module 20 for Critical/Low; suppress for Overstock.
- Review thresholds quarterly or after usage trends change.
- Integrate with Module 19 (Expiration) and Module 20 (Ordering).

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Module 19

PROCESS DOCUMENTATION

1. Information	
Process Name:	Expiration Date Monitoring Process
Process Owner:	Clinic Doctor / Clinic Owner
Department/Unit:	Inventory, Supplies & Operations
Document Prepared By:	Tristan Paolo C. De Castro
Date Prepared:	September 12, 2026
Version Number:	1.0
Last Updated:	September 12, 2026

2. Overview

Purpose: To automate the expiration-date checks the clinic already does manually.

Process Description: The system tracks expiration dates per supply batch, compares against the current date, and alerts the doctor a set number of days before expiry.

Actors Involved:
- System — tracks, compares, alerts
- Doctor — receives alert

3. Inputs

- Expiration Date per Batch - Supply entry
- Alert Window (days before expiry) - System config

4. Activities

1. System tracks expiration dates per batch - System
2. System compares against current date - System
3. System alerts doctor a set number of days before expiry - System
4. Alert appears on dashboard - System

5. Outputs

Expiration Alert -> Doctor (Module 21: Admin & Operations Dashboard)

6. Stakeholders Analysis

Internal:
- Doctor / Clinic Owner — receives alerts
- IT Department — maintains monitoring logic

External:
- None

Note: *No receptionist is involved. Alerts are automatic.*

7. Flow Summary

Start → Track Expiration Dates → Compare to Today → Near Expiry? → Yes → Alert Doctor → End

8. Performance Metrics	
Expiration checks	Manual, periodic (Target: Automatic, continuous)
Expired supplies used	Occasional (Target: 0%)
Alert timing	Late (Target: In advance)

9. Issues and Challenges

- Manual expiration checks are periodic
- Expired supplies may be used unknowingly
- No advance warning

10. Improvement Opportunities

- Auto-tracking
- Auto-alerts
- Dashboard visibility - with Module 21 (Admin & Operations Dashboard)

11. Automation Potential

- Tracking - Automatic
- Comparison - Automatic
- Alerting- Automatic

Only remaining human touchpoint: Doctor acts on alert.

12. Business Process Lifecycle Analysis	
Design	Define expiration alert windows.
Execution	System tracks and alerts automatically.
Monitoring	Track expired items and alert accuracy.
Analysis	Identify commonly expiring items.
Optimization	Adjust order quantities to reduce waste.

13. Evaluation and Recommendation

Module 19 automates what the clinic already does manually — with continuous monitoring instead of periodic checks.

Recommendation:

- Implement Module 19 after Module 17 (Supply Inventory Tracking) — needs supply data.
- Set an alert window at 30–60 days — enough time to use or reorder.
- Auto-flag on dashboard — immediate visibility

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Module 20

Module 20: Supply Ordering & Audit Process

Revision Note: This version updates Module 20 to make ordering **fully automatic**. When Module 18 triggers a **Critical Low** or **Low** stock alert, the system automatically drafts and sends the Purchase Order directly to the designated supplier — **no manual confirmation required**. Orders are batched and dispatched **at the end of each day** (configurable). **Overstock** alerts suppress ordering entirely. Module 22 tracks order status, and Module 23 provides supplier routing and performance feedback.

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1. Information

Field	Details
Process Name	Supply Ordering & Audit Process
Process Owner	Clinic Doctor / Clinic Owner
Department/Unit	Inventory, Supplies & Operations
Document Prepared By	Tristan Paolo C. De Castro
Date Prepared	September 12, 2026
Version Number	3.0
Last Updated	September 25, 2026

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2. Overview

Purpose: To fully automate supply ordering and auditing — removing the doctor's manual approval step — while respecting critical, low, and overstock thresholds. Orders are sent directly to suppliers without human intervention.

Process Description: When Module 18 flags **Critical Low** or **Low** stock, the system automatically drafts a restock order using supplier data from Module 23. At the **end of each day** (or immediately for critical items, if configured), the system generates a structured PDF Purchase Order and emails it directly to the designated supplier. **No doctor approval is required.** Module 22 then tracks the order from submission to delivery. If Module 18 flags **Overstock**, Module 20 suppresses ordering entirely and logs a review note for the doctor. All orders are logged, and periodic audit reports are generated automatically.

Actors Involved:

- **System** – drafts orders, selects suppliers, sends POs, logs transactions, generates audits
- **Doctor** – receives notifications, reviews audit reports, handles exceptions only
- **Supplier** – receives and fulfills orders automatically
- **Module 22** – tracks order status after submission
- **Module 23** – provides supplier records, backup suppliers, and routing data

3. Inputs

- **Critical Low-Stock Alert** – Module 18
- **Low-Stock Alert** – Module 18
- **Overstock Alert** – Module 18 (for suppression only)
- **Live Stock Quantity** – Module 17
- **Expiration Data** – Module 19
- **Supplier Info** – Module 23: primary supplier, backup supplier, representative, email, phone, lead time, payment terms
- **Order History** – Module 20
- **Order Status** – Module 22

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4. Activities

1. System receives threshold alert from Module 18 — **System**
2. System classifies alert as **Critical Low**, **Low**, or **Overstock** — **System**
3. If **Critical Low** or **Low**, system retrieves supplier mapping from Module 23 — **System**
4. System auto-drafts restock order with item, current stock, suggested quantity, primary supplier, and backup supplier — **System**
5. System queues order for **end-of-day batch dispatch** (or immediate if critical and configured) — **System**
6. If **Overstock**, system suppresses auto-draft and logs a review note for the doctor — **System**
7. At scheduled dispatch time, system generates a structured PDF Purchase Order — **System**
8. System emails the PDF PO directly to the designated supplier — **System**
9. System logs order (what, when, from whom, by whom) — **System**
10. System passes the order to Module 22 for status tracking — **System**
11. Module 22 tracks order confirmation and delivery — **System / Module 22**
12. System periodically generates an audit report — **System**
13. System feeds supplier performance data back to Module 23 — **System**
14. Doctor receives notification of orders sent and any exceptions — **Doctor**

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5. Outputs

- **Auto-Drafted Order** → System (internal)
- **Sent Purchase Order** → Supplier
- **Order Confirmation Log** → Module 20
- **Supplier Audit Trail** → Module 20 / Doctor
- **Periodic Audit Report** → Doctor
- **Order Status Record** → Module 22
- **Supplier Performance Data** → Module 23
- **Overstock Review Note** → Doctor (Module 21: Admin & Operations Dashboard)
- **Exception Alerts** (e.g., email bounce) → Doctor

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6. Stakeholders Analysis

Internal:

- **Doctor / Clinic Owner** – receives notifications, reviews audits, handles exceptions
- **IT Department** – maintains ordering system, supplier routing, and audit logic

External:

- **Suppliers** – receive and fulfill orders automatically
- **Affiliate Companies** – linked suppliers under same parent or partner network
- **Backup Suppliers** – activated automatically when primary supplier is unavailable

Note: No receptionist is involved. Ordering is fully automatic; the doctor is informed, not asked.

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7. Flow Summary

Start → Threshold Alert from Module 18 → Classify Alert → Critical Low or Low? → Yes → Retrieve Supplier from Module 23 → Auto-Draft Order → Queue for End-of-Day Dispatch → Generate PDF PO → Email Supplier → Module 22 Tracks Status → Track Delivery → Auto-Audit Report → Update Module 23 Performance → End

Alternative paths:

- **Overstock Alert** → No auto-draft → Log review note for doctor → End
- Email bounce → System alerts doctor for manual follow-up
- Primary supplier unavailable → Module 23 provides backup supplier automatically
- Critical Low (if configured) → Immediate dispatch instead of end-of-day batch

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8. Performance Metrics

Metric	Current	Target
Ordering time	Manual, ad hoc	Fully automatic
Audit time	Hours, manual	Zero (auto)
Order accuracy	Human error	100%
Audit trail	None	Full
Critical alert response	Not tracked	Same-day dispatch
Supplier routing accuracy	Manual	Automatic via Module 23
Overstock suppression	None	Automatic
Order visibility	None	Real-time via Module 22
Manual approvals required	Every order	Zero

9. Issues and Challenges

- Ordering is reactive and manual
- Auditing supplies is time-consuming
- No audit trail
- No systematic reorder timing
- No distinction between critical and low stock in ordering
- No overstock control
- No centralized supplier data
- No order status visibility after submission
- Manual approval creates delays and bottlenecks

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10. Improvement Opportunities

- Fully automatic order drafting and dispatch
- End-of-day batch processing to consolidate orders
- Immediate dispatch for critical items (configurable)
- Overstock suppression
- Full audit trail
- Auto-generated audit report
- Supplier routing via Module 23
- Order tracking via Module 22
- Supplier performance feedback to Module 23
- Dashboard visibility via Module 21

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11. Automation Potential

- Order drafting – **Automatic**
- Supplier selection – **Automatic** from Module 23
- Order dispatch – **Automatic** (end-of-day batch or immediate for critical)
- Logging – **Automatic**
- Audit report – **Automatic**
- Order status tracking – **Automatic** via Module 22

Only remaining human touchpoint: Doctor reviews audit reports and handles exceptions (e.g., email bounce, overstock review).

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12. Business Process Lifecycle Analysis

Phase	Description

Design	Create auto-drafting rules per threshold, supplier routing rules, overstock suppression, end-of-day batch schedule, and audit report rules
Execution	System drafts, dispatches, logs, and audits automatically; Module 22 tracks status
Monitoring	Track order frequency, delivery times, audit accuracy, supplier performance, overstock incidents
Analysis	Identify supplier performance and usage trends
Optimization	Adjust reorder points, thresholds, suppliers, batch timing, and order quantities

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13. Evaluation and Recommendation

Module 20 now removes the doctor entirely from the ordering loop. Critical Low and Low alerts trigger automatic ordering, with dispatch batched at the end of each day. Overstock alerts suppress ordering. Module 23 ensures correct supplier routing, and Module 22 provides full order-status visibility. The doctor is notified, not asked.

Recommendations:

- Implement **Module 23 (Supplier Management)** before Module 20 — Module 20 needs supplier data to auto-draft and route orders.
- Implement **Module 18 (Low-Stock, Critical Low-Stock, and Overstock Alerts)** before Module 20 — Module 20 depends on threshold alerts.
- Implement Module 20 alongside **Module 22 (Order Monitoring)** — they work together for end-to-end order lifecycle visibility.
- Pre-load supplier info, including primary and backup suppliers, so drafts are ready to send.
- Configure end-of-day batch time (e.g., 6:00 PM) for non-critical orders.
- Optionally enable immediate dispatch for Critical Low items.
- Enable overstock suppression to prevent unnecessary purchasing.
- Auto-email audit reports monthly — zero effort.
- Review thresholds and supplier performance quarterly.
- Monitor for email bounce exceptions and set up fallback notification (e.g., SMS or dashboard alert).

Module 22

Module 22: Order Monitoring Process

1. Information

Field	Details
Process Name	Order Monitoring Process
Process Owner	Clinic Doctor / Clinic Owner
Department/Unit	Inventory, Supplies & Operations
Document Prepared By	Tristan Paolo C. De Castro
Date Prepared	September 25, 2026
Version Number	2.0
Last Updated	September 25, 2026

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2. Overview

Purpose: To give the doctor a single, real-time view of every order generated by Module 20 — from automatic dispatch to delivery — so nothing is lost, delayed, forgotten, or double-ordered.

Process Description: The system tracks every Purchase Order created and dispatched by Module 20 through defined lifecycle stages. Since ordering is now fully automatic (no approval step), orders begin at **Queued for Dispatch**, then move through **Sent → Confirmed → In Transit → Delivered → Completed** (or **Cancelled/Partially Delivered**). The doctor views all orders on a monitoring dashboard, filters by status, supplier, or date, and acts only on stalled or overdue orders. The system auto-flags orders that exceed expected lead times.

Actors Involved:

- **System** – tracks order status, updates stages, flags delays, notifies
- **Doctor** – views orders, updates statuses when manual confirmation is needed, acts on delays
- **Supplier** – receives order, confirms, delivers (external)

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3. Inputs

- **Auto-Drafted Order** – Module 20 (Supply Ordering & Audit)
- **Dispatched Order** – Module 20
- **Supplier Confirmation** – External (email or manual entry)
- **Delivery Receipt / Confirmation** – External (manual entry or supplier)
- **Expected Lead Time** – Module 23 (Supplier Management)
- **Order History** – Module 20

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4. Activities

1. System receives new order from Module 20 — **System**
2. System assigns initial status: **Queued for Dispatch — System**
3. System dispatches order at end-of-day batch (or immediately for Critical Low) → status becomes **Sent / Submitted — System**
4. Doctor or system logs supplier confirmation → status becomes **Confirmed — Doctor / System**
5. Supplier dispatches order → status becomes **In Transit — Doctor / System**
6. Doctor logs delivery or system receives delivery confirmation → status becomes **Delivered — Doctor / System**
7. Doctor verifies stock received → status becomes **Completed — Doctor**
8. System retains full order timeline and history **— System**
9. Doctor views all orders on monitoring dashboard **— Doctor**
10. System flags orders exceeding expected lead time as **Overdue — System**
11. System notifies doctor of overdue or stalled orders **— System**

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5. Outputs

- **Order Monitoring Dashboard** → Doctor (Module 21: Admin & Operations Dashboard)
- **Order Status Timeline** → Doctor
- **Overdue / Stalled Order Alerts** → Doctor (Module 21)
- **Order History Log** → Module 20
- **Supplier Performance Data** → Module 23 (Supplier Management)

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6. Stakeholders Analysis

Internal:

- **Doctor / Clinic Owner** – views orders, updates statuses, acts on delays
- **IT Department** – maintains tracking logic and status rules

External:

- **Suppliers** – receive orders, confirm, dispatch, deliver

Note: No receptionist is involved. Order tracking is automatic with doctor oversight. No approval step exists — orders are dispatched automatically by Module 20.

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7. Flow Summary

Start → Order Auto-Created by Module 20 → Queued for Dispatch → Sent to Supplier → Supplier Confirms → In Transit → Delivered → Doctor Verifies → Completed → End

Alternative paths:

- No supplier response within lead time → **Stalled Order Alert**
- Delivery exceeds expected date → **Overdue Alert**
- Partial delivery → status remains **Partially Delivered**, re-flagged
- Order cancelled by supplier → status becomes **Cancelled**, doctor notified

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8. Performance Metrics

Metric	Current	Target
Order visibility	Scattered / manual	Single dashboard
Status tracking	None	Real-time
Lost / delayed orders	Common	Rare
Order follow-up time	Manual, repeated	One-click view
Order history	None	Full timeline
Overdue detection	Manual	Automatic
Approval bottleneck	Every order	Zero

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9. Issues and Challenges

- No visibility on order status after submission to supplier
- No tracking of delivery timelines

- No way to confirm if supplier received the order
- No history of past orders
- Manual follow-up required for every order
- Risk of duplicate orders when status is unknown

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10. Improvement Opportunities

- Centralized order monitoring dashboard
- Status-based filtering and search
- Overdue and stalled order alerts
- Full order timeline and audit trail
- Integration with Module 20 (Ordering) and Module 21 (Dashboard)
- Supplier performance feedback to Module 23

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11. Automation Potential

- Status tracking – **Automatic**
- Dashboard aggregation – **Automatic**
- Overdue flagging – **Automatic**
- Notifications – **Automatic**
- History logging – **Automatic**

Only remaining human touchpoint: Doctor views orders and updates status when manual confirmation is needed.

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12. Business Process Lifecycle Analysis

Phase	Description
Design	Define order status stages and tracking rules
Execution	System tracks orders; doctor monitors and updates
Monitoring	Track order frequency, delivery times, stalled orders
Analysis	Identify supplier performance and bottlenecks
Optimization	Adjust supplier selection and reorder timing

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13. Evaluation and Recommendation

Module 22 gives the doctor full visibility over every order from automatic dispatch to delivery — eliminating guesswork, manual follow-up, and duplicate ordering. It works directly with **Module 20 (Supply Ordering & Audit)** and **Module 21 (Admin & Operations Dashboard)**. Since Module 20 now dispatches automatically, Module 22 is the doctor's primary window into the procurement cycle.

Recommendations:

- Implement Module 22 alongside Module 20 — they work together.
- Define clear status stages: **Queued for Dispatch, Sent, Confirmed, In Transit, Delivered, Completed, Cancelled, Partially Delivered**.
- Auto-flag orders that remain "Sent" beyond expected lead time.
- Show order monitoring on the main dashboard — no separate login.
- Allow manual status updates for offline supplier confirmations.
- Feed delivery performance data to Module 23 (Supplier Management).

Module 23

Module 23: Supplier Management Process

1. Information

Field	Details
Process Name	Supplier Management Process
Process Owner	Clinic Doctor / Clinic Owner
Department/Unit	Inventory, Supplies & Operations
Document Prepared By	Tristan Paolo C. De Castro
Date Prepared	September 25, 2026
Version Number	1.1
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2. Overview

Purpose: To centralize all supplier information — affiliate companies, representatives, contact details, product mappings, and backup suppliers — so ordering is fast, reliable, and resilient, with automatic routing to backup suppliers when primary suppliers are unavailable.

Process Description: The system maintains a master supplier database. Each supplier record includes company name, contact person/representative, phone, email, address, products supplied, lead time, payment terms, and status (Active / Inactive / Backup). Primary and backup suppliers are linked to specific inventory items so Module 20 can auto-route orders. When a primary supplier is unavailable, the system automatically routes to the backup supplier. The doctor manages supplier records and reviews performance data.

Actors Involved:

- **Doctor** – adds, edits, views supplier records; assigns primary/backup suppliers
- **System** – stores, links, retrieves supplier data; auto-routes orders; activates backup suppliers
- **Suppliers** – external parties whose details are recorded

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3. Inputs

- **Supplier Information** – Doctor (manual entry): company name, representative, contact details, address
- **Product–Supplier Mapping** – Doctor (which items each supplier provides)
- **Lead Time & Payment Terms** – Doctor
- **Backup Supplier Assignments** – Doctor
- **Order History** – Module 20 (Supply Ordering & Audit)
- **Supplier Performance Data** – Module 22 (Order Monitoring)

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4. Activities

1. Doctor adds new supplier record — **Doctor**
2. System stores company name, representative, contact details, products, lead time, payment terms — **System**
3. Doctor links suppliers to inventory items (primary and backup) — **Doctor**
4. System uses supplier data to auto-draft and route orders in Module 20 — **System**
5. When primary supplier is unavailable, system **automatically routes to backup supplier** — **System**
6. Doctor is notified of automatic backup routing — **System / Doctor**
7. System tracks supplier performance from order history — **System**
8. Doctor views supplier list and performance on dashboard — **Doctor**
9. Doctor updates supplier details as needed — **Doctor**
10. System logs all supplier changes in audit trail — **System**

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5. Outputs

- **Supplier Master List** → Doctor
- **Supplier–Item Mapping** → Module 20 (Supply Ordering & Audit)
- **Supplier Performance Report** → Doctor (Module 21: Admin & Operations Dashboard)
- **Automatic Backup Routing** → Module 20
- **Supplier Contact Directory** → Doctor
- **Order Routing Rules** → Module 20

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6. Stakeholders Analysis

Internal:

- **Doctor / Clinic Owner** – manages supplier records, reviews performance
- **IT Department** – maintains supplier database and routing logic

External:

- **Suppliers** – primary and backup, receive orders, provide goods
- **Affiliate Companies** – linked suppliers under same parent or partner network

Note: No receptionist is involved. Supplier data is centralized and maintained by the doctor. Backup routing is automatic.

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7. Flow Summary

Start → Doctor Adds Supplier → System Stores Record → Doctor Links to Items → Module 20 Uses Supplier Data → Orders Routed → Performance Tracked → Doctor Reviews → End

Alternative paths:

- Primary supplier unavailable → **System automatically routes to Backup Supplier** → Doctor notified
- Supplier performance poor → Doctor reassigns items to alternative supplier
- New supplier added → Record created, linked to items, available for ordering

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8. Performance Metrics

Metric	Current	Target
Supplier info	Scattered / manual	Centralized
Contact lookup time	Minutes	Seconds
Backup supplier readiness	None	Pre-linked, auto-activated
Order routing accuracy	Manual	Automatic
Supplier performance visibility	None	Full history
Supplier data maintenance	Ad hoc	Structured, audited

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9. Issues and Challenges

- Supplier information scattered across phone contacts, messages, and paper
- No backup supplier list

- No performance tracking
- Manual lookup during ordering
- No centralized contact directory
- No way to compare supplier reliability or pricing
- Manual backup activation causes delays

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10. Improvement Opportunities

- Centralized supplier database
- Primary/backup supplier linking per item
- Auto-routing of orders to correct supplier
- **Automatic backup activation** when primary unavailable
- Supplier performance tracking from Module 22
- One-click supplier contact and order history
- Affiliate company grouping for bulk negotiations

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11. Automation Potential

- Storage – **Automatic**
- Retrieval – **Automatic**
- Order routing – **Automatic**
- Backup activation – **Automatic**
- Performance tracking – **Automatic**
- Contact directory – **Automatic**

Only remaining human touchpoint: Doctor adds and updates supplier records, and reviews performance.

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12. Business Process Lifecycle Analysis

Phase	Description
Design	Create supplier master list with primary/backup mapping and auto-routing rules
Execution	System stores, links, routes, and auto-activates backups
Monitoring	Track supplier performance and order fulfillment

Analysis	Identify reliable vs. unreliable suppliers
Optimization	Reassign items, negotiate terms, update backup list

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13. Evaluation and Recommendation

Module 23 eliminates the need to search for supplier contacts and ensures orders always have a destination — even when a primary supplier is unavailable, since the system now routes to backup suppliers automatically. It works with **Module 20 (Supply Ordering & Audit)** and **Module 22 (Order Monitoring)** to create a complete procurement ecosystem.

Recommendations:

- Implement Module 23 before Module 20 — Module 20 needs supplier data to auto-draft and route orders.
- Pre-load all suppliers, including backups and affiliate companies.
- Link every inventory item to at least one primary and one backup supplier.
- **Configure automatic backup routing rules** — no manual activation required.
- Track supplier performance automatically using Module 22 data.
- Review supplier list and terms quarterly.
- Group affiliate companies for potential bulk ordering or shared terms.

Module 21

Module 21: Doctor's Daily Workflow Process

1. Information

Field	Details
Process Name	Doctor's Daily Workflow Process
Process Owner	Clinic Doctor / Clinic Owner
Department/Unit	Admin & Operations Dashboard
Document Prepared By	Tristan Paolo C. De Castro
Date Prepared	September 12, 2026
Version Number	2.0
Last Updated	September 25, 2026

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2. Overview

Purpose: To give the doctor one control-center view tying every other module together — now including automatic order notifications and real-time order status — directly addressing the daily-repeated-task and pain-point questions from the interview.

Process Description: The doctor logs into the dashboard and sees today's appointments, pending recalls, overdue balances, low-stock items, automatically sent orders, order statuses, and daily income in one screen. Since ordering is now fully automatic, the doctor no longer approves restock orders — they review notifications, monitor order status, and handle only exceptions (email bounces, overstock reviews, delayed deliveries). All other actions remain one-click.

Actors Involved:

- **Doctor** – views dashboard, monitors orders, handles exceptions, takes one-click actions
- **System** – aggregates data from all modules

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3. Inputs

- **Today's Appointments** – Module 4 (Schedule & Conflict Management)
- **Pending Recalls** – Module 5 (Patient Registration & Authentication)
- **Overdue Balances** – Module 15 (Outstanding Balance Monitoring)
- **Low-Stock / Critical / Overstock Alerts** – Module 18 (Low-Stock, Critical Low-Stock, and Overstock Alerts)
- **Sent Orders / Order Notifications** – Module 20 (Supply Ordering & Audit)
- **Order Status** – Module 22 (Order Monitoring)
- **Overstock Review Notes** – Module 20

- **Daily Income** – Module 16 (Income & Expense Reporting)

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4. Activities

1. Doctor logs into dashboard — **Doctor**
2. Dashboard shows all key metrics — **System**
3. Doctor reviews flagged items (Critical/Low stock, Overstock notes) — **Doctor**
4. Doctor monitors automatically sent orders and their statuses — **Doctor**
5. Doctor handles exceptions (email bounce, delayed delivery, overstock review) — **Doctor**
6. Doctor assigns walk-in slots (one-click) — **Doctor**
7. Doctor charts + taps supplies used (during visits) — **Doctor**
8. Doctor views end-of-day income summary — **Doctor**

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5. Outputs

- **Completed Daily Workflow** → Doctor
- **Exception Handling** → Module 20 / Module 22
- **Overstock Review** → Module 20
- **Walk-in Assignments** → Module 4
- **Chart Entries** → Module 9 & Module 17
- **Daily Income View** → Doctor

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6. Stakeholders Analysis

Internal:

- **Doctor / Clinic Owner** – primary user
- **IT Department** – maintains dashboard

External:

- None directly (patients benefit indirectly)

Note: No receptionist is involved. Dashboard replaces the receptionist's coordination role and removes the doctor's manual approval duties.

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7. Flow Summary

Start → Log In → View Dashboard → Review Flagged Items / Sent Orders / Order Status → One-Click Actions (exceptions only) → Chart During Visits → View Income → End

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8. Performance Metrics

Metric	Current	Target
Non-clinical time per day	1–2 hours	Under 15 minutes
Dashboard views	Multiple (calendar, chart, supplies)	Single screen
Manual data entry	Frequent	Zero
Manual order approvals	Every order	Zero
Income visibility	End of day, manual	Real-time
Order status visibility	None	Real-time

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9. Issues and Challenges

- Doctor checks calendar, chart shelf, supply list separately
- Manual income computation
- No unified view
- Constant context-switching
- Previously required manual order approvals (now removed)

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10. Improvement Opportunities

- Single dashboard
- One-click actions (exceptions only)
- Real-time metrics
- Zero data entry
- Automatic order notifications
- Real-time order status monitoring

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11. Automation Potential

- Data aggregation – **Automatic** from all modules
- Order dispatch – **Automatic** (Module 20)
- Order status – **Automatic** (Module 22)
- Reporting – **Automatic**

Only remaining human touchpoint: Doctor views, monitors exceptions, and clicks for non-ordering actions.

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12. Business Process Lifecycle Analysis

Phase	Description
Design	Create a single dashboard aggregating all modules
Execution	Doctor views, monitors, and handles exceptions with one click
Monitoring	Track dashboard usage and action frequency
Analysis	Identify most-used features and pain points
Optimization	Reorder/redesign based on usage

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13. Evaluation and Recommendation

Module 21 is the single biggest time-saver for the doctor — it replaces 4–5 separate manual checks with one screen and one-click actions. With automatic ordering now in place, the doctor no longer approves restock orders, further reducing daily workload.

Recommendations:

- Implement Module 21 last — needs data from all other modules.
- Design for one-screen visibility — no scrolling.
- Prioritize one-click actions — every approval should be one tap.
- Display automatic order notifications and order status prominently.
- Surface only exceptions (bounces, delays, overstock) as actionable items.